

Homework (Jalapeno)

Q1. Simplify $\sqrt{16}$.

- (A) 2
- (B) 4
- (C) 6
- (D) 8
- (E) 10

Q2. The ancient Babylonians used $\sqrt{9}$ to represent the side length of a square with area 9. What is the value of $\sqrt{9}$?

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Q3. Simplify $\sqrt[3]{27}$.

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Q4. The ancient Egyptians used $\sqrt[3]{64}$ to calculate volumes of cubes. What is the value of $\sqrt[3]{64}$?

- (A) 3
- (B) 4
- (C) 5
- (D) 6
- (E) 7

Q5. Convert $\sqrt{25}$ to rational exponent form.

- (A) $25^{1/4}$
- (B) $25^{1/2}$
- (C) $25^{3/4}$
- (D) $25^{3/2}$
- (E) $25^{2/3}$

Q6. Simplify $\sqrt[3]{8}$.

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 5

Q7. The ancient Greeks used $\sqrt{36}$ to calculate distances. What is the value of $\sqrt{36}$?

- (A) 5
- (B) 6
- (C) 7
- (D) 8
- (E) 9

Q8. Convert $\sqrt[4]{81}$ to rational exponent form.

- (A) $81^{1/3}$
- (B) $81^{1/4}$
- (C) $81^{2/3}$
- (D) $81^{3/4}$
- (E) $81^{1/2}$

Open Ended Questions (FRQ)

Q9. Simplify $\sqrt{100} + \sqrt[3]{1000}$. Show your work and calculate the final value.

Q10. The ancient Chinese used $\sqrt{49}$ to represent the side length of a square with area 49. Convert $\sqrt{49}$ to rational exponent form and calculate its value. Show your work and explain your reasoning.

Chef's Tips

- Tip 1: Remind students to simplify radicals by finding perfect squares or cubes.
- Tip 2: Encourage students to convert between radical and rational exponent form to check their work.
- Tip 3: Have students verify their answers using different methods or approaches.